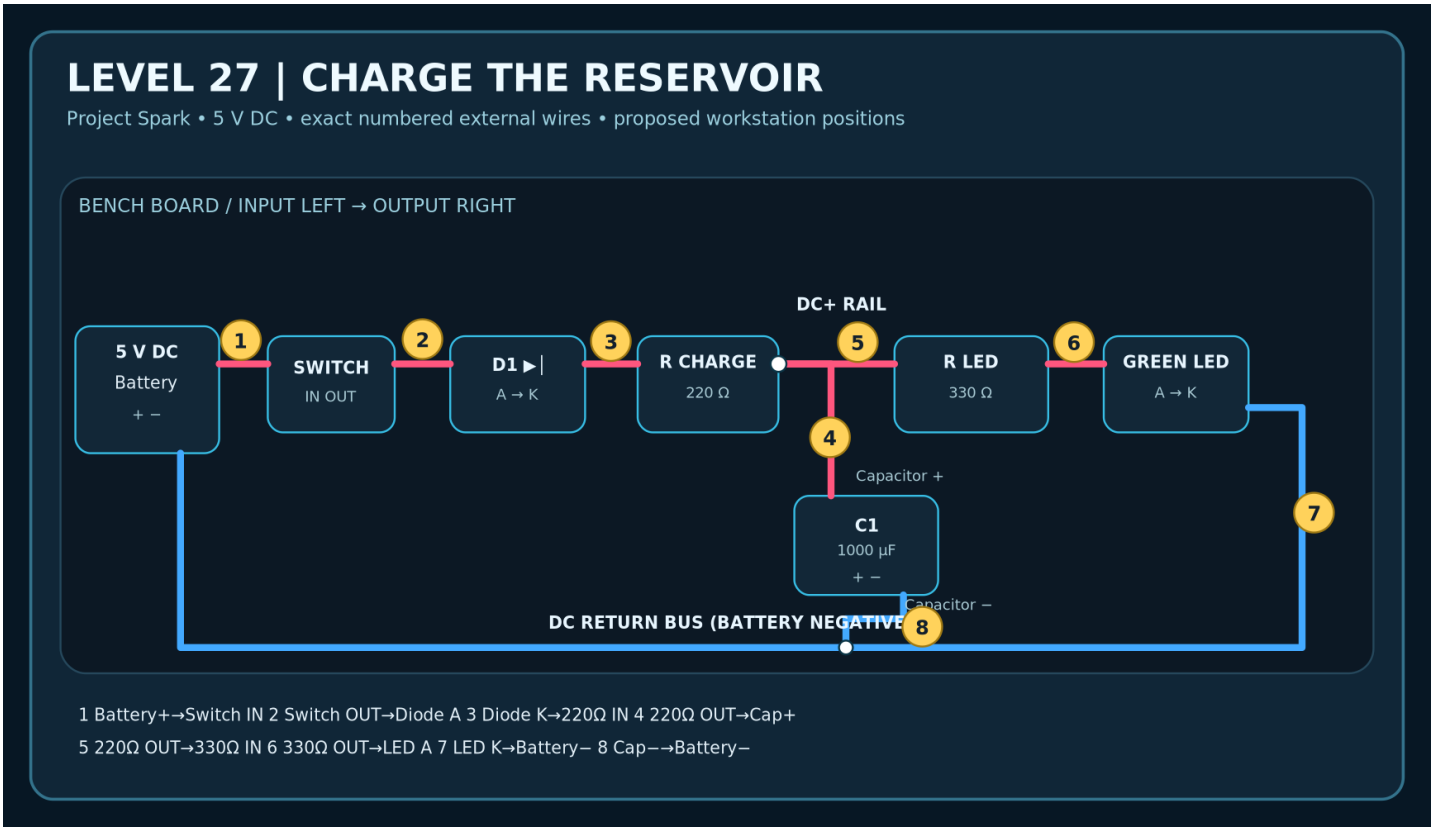
LEVEL 27 • CHARGE THE RESERVOIR

Project Spark | Creation process | 5 V DC capacitor energy-storage demonstration

This is a DC-compatible reservoir lesson, not an AC half-wave ripple measurement. The uploaded Level 27 definition still contained Level 26 bridge objectives, which have been replaced in the included editable TXT.



Eight numbered external connections

#	From	To
1	Battery_positive	SwitchInput
2	SwitchOutput	D_ISOLATE_Anode
3	D_ISOLATE_Cathode	R_CHARGE_Input
4	R_CHARGE_Output	C_RESERVOIR_Positive
5	R_CHARGE_Output	R_LED_Input
6	R_LED_Output	LED_Anode
7	LED_Cathode	Battery_negative
8	C_RESERVOIR_Negative	Battery_negative

Configuration and target ranges

Slot_01: diode D_ISOLATE • Slot_02: 220 Ω R_CHARGE • Slot_03: 1000 µF C_RESERVOIR • Slot_04: 330 Ω R_LED • fixed green LED and 5 V source.

Runtime bindings

Slot	Component ID	Terminal A	Terminal B
Slot 01	D_ISOLATE_Component	D_ISOLATE_Anode	D_ISOLATE_Cathode
Slot 02	R_CHARGE_Component	R_CHARGE_Input	R_CHARGE_Output
Slot 03	C_RESERVOIR_Component	C_RESERVOIR_Positive	C_RESERVOIR_Negative
Slot 04	R_LED_Component	R_LED_Input	R_LED_Output

Six completion objectives

1. Supply voltage 4.8–5.2 V
2. Forward diode drop 0.45–1.0 V
3. Charged capacitor voltage 3.1–3.7 V
4. Green LED illuminated
5. LED branch resistor current 3–6 mA
6. Charge resistor voltage 0.7–1.2 V

Important capacitor Smart Slot note

The existing Smart Slot Bridge uses _Input/_Output matching for non-LED categories. Capacitors typically have differently named positive/negative terminals. For Slot_03, use a legacy rule with exact physical terminal names and Smart Binding disabled on that slot, or implement a capacitor-specific runtime terminal matcher. Keep the overall bridge Use Smart Slots setting enabled.

Editor and transient requirements

Import Level_27_FIXED.txt through the existing editable ScriptableObject importer, not as a native Unity .asset. Put SparkLevel27ReservoirDriver only on the Level 27 layout, register the existing capacitor solver definition, and test in Play Mode. The creator script clones a working drawing template but cannot author unknown page fields without their C# schema.